

An Autonomous End-to-End Agent for Battery Cell Design with Verifiable Outcomes

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Abstract

We report the first end-to-end battery design agent for full-chain cell design: given nothing but a natural-language contract, a single autonomous loop traverses the whole chain of cell-design levers—material system, formulation, electrode architecture, thermal management, safety, and aging—and returns a complete adjudicated design together with a multi-level true-compute endorsement (ML potentials, two independent DFT codes, and a classical-force-field diffusion simulation). Its steering logic is a transferable mechanism: cross-scale bottleneck routing (failures diagnosed and routed to the scale of their cause), a ceiling assessment that triggers material-level search only when the current system provably cannot close the gap, and code-computed verdicts for every criterion with audit-chain rejection. Eight scenario tasks with absolute numeric criteria are given, under matched resources (same model, same tools, same task texts, each condition’s standard budget), to three decision-makers: the governed agent, a protocol-free agent on the same harness, and a Bayesian optimizer configured to the standard of the strongest comparable published agent framework. The governed agent attains all eight contracts under two DeepSeek-generation models, and four of eight under each of two foreign-vendor models; the eight under the primary model are

replicated under three seeds (24/24 attained). The optimizer attains three, exactly the three whose bottleneck is structural. The protocol-free agent attains none within the same measurement space: of its seven self-reported successes, two sit on a lithium-metal cell outside the validated parameter sets and five dissolve under contract-caliber re-adjudication. Ablation measures each governance component in turn: ceiling assessment with material escalation is causal for two of three material-bottleneck tasks, forced architecture exploration for one of six, and the molecular funnel’s three-model vote was exercised but never verdict-changing in this suite. The simulator is anchored to a production cell: the beginning-of-life 0.1C discharge of a commercial LG M50T matches the calibrated parameter set to 0.45% capacity and 20.6 mV RMSE. The true-compute endorsement of the flagship finalists confirms the screening verdicts in two independent codes, and a 1 ns classical-force-field diffusion simulation reports $D_{\text{Li}} = 5.4 \times 10^{-11} \text{ m}^2/\text{s}$. Protocol, simulation library, adjudication layer, and experiment harnesses are open source at <https://github.com/wangyuanxty/BatteryDesignAgent>.
Keywords: LLM agents, battery cell design, cross-scale bottleneck routing, multi-fidelity endorsement, mechanized adjudication, Bayesian optimization

1. Introduction

Battery development is a key link in transport electrification and grid energy storage, yet the field itself is constrained by a core dilemma: *design space versus evaluation cost*. Performance goals (energy density, rate capability, cycle life, safety, cost) are mutually coupled; design variables span material systems (active materials, electrolyte formulations), electrode microstructure (particle radius, porosity, tortuosity), and operating conditions, together forming a high-dimensional nonlinear search space. Mean-

while, evaluating any candidate design is expensive: cycle-life assessment commonly requires thousands of hours of continuous charge–discharge testing [1], and key microscopic parameters (solid/liquid transport coefficients, reaction-rate constants, porosity, initial lithium concentration) are difficult or impossible to measure directly [1] and must be inferred from macroscopic voltage/capacity trajectories, the well-known *inverse problem* [1]. Under this constraint the classical trial-and-error paradigm iterates slowly: battery design is therefore an *evaluation-constrained* problem.

Figure 1 previews the study: the application settings that motivate the suite (1a), the governance diagnostic that defines our contribution (1b), and the architecture (1c).

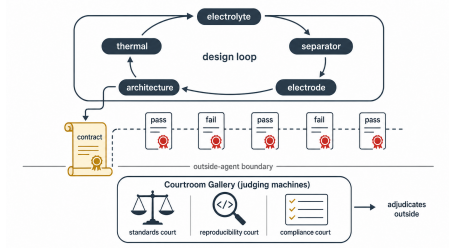
1.1. From Black-Box Search to Agentic Design

Early data-driven methods formalized the problem as *black-box optimization* (BBO): numerical algorithms such as Bayesian optimization [2] or genetic algorithms [3] iteratively query a physics simulator (the battery digital twin, e.g. the Doyle-Fuller-Newman model in the open-source platform PyBaMM [4]) to minimize the misfit between simulated and observed trajectories. Such methods are general and carry theoretical convergence guarantees, but they are inherently blind: the simulator is treated as an opaque black box, physical intuition and literature knowledge cannot be exploited, sample efficiency is low, and they frequently converge to physically implausible optima (the multiple parameter sets that reproduce near-identical macroscopic outputs, the *equifinality* phenomenon) [5].

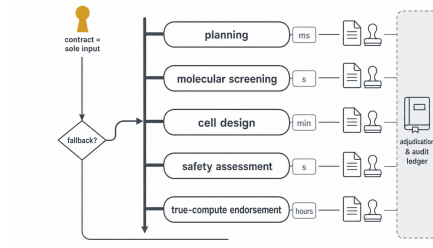
The rise of large language models (LLMs) has driven a new *agentic science* paradigm (LLM as the reasoning engine, external tools and knowledge bases as its hands) which has unfolded in battery research across three lay-



(a) Deployment scenes: fast-charge transport (left), grid storage (center), and drone powertrains (right)—the application settings of the eight scenario contracts.



(b) The governance idea: the design loop over all five lever families ends at the contract, and every verdict is issued by a judging layer outside the agent.



(c) Architecture at a glance: five fidelity rungs, each gate kept by one entry: computational cost rises left to right.

Figure 1: Overview of the study. (a) The application settings that motivate the eight scenario contracts. (b) The central diagnostic: a design loop that ends at the contract and an adjudication layer that stands outside it. (c) The multi-precision architecture: cheap proxies first, true first principles last.

ers. At the *knowledge layer*, RAG research assistants automate literature synthesis: Words-to-Watts proposed the BatteryGPT concept [6]; ChatSSB

integrates a literature corpus with an expert knowledge base for solid-state battery research (GPT-o1 judged 8.77/10) [7]; and the top-level review provides the three-layer “database–model–task” roadmap [8]. At the *computation layer*, LiPM injects physical laws (charge conservation) into the pre-training objective of a temporal foundation model, reducing mean squared error by an average 39.2% over state-of-the-art baselines across 62 downstream tasks [9]; Battery-Sim-Agent deployed an LLM agent in the simulator-in-the-loop on a high-fidelity PyBaMM twin, reducing curve-matching error by 67–95% (up to 58–97% MAPE in routine mode) against black-box optimization through a three-step reasoning loop of multimodal feedback, explicit physical hypotheses, and structured parameter updates, validated on long-horizon degradation fits and real CALCE/NASA data [5]. At the *experiment layer*, a chained human-in-the-loop framework for four-electron Zn-I2 electrolyte design, driven by GPT-4 literature mining and mechanism summarization, recommended [EMIM]Cl—verified wet-lab to cycle above 1000 times at high rate with Coulombic efficiency above 99.3% [10]; ChatBattery further runs an end-to-end pipeline (hypothesis generation → database retrieval → MACE-MP machine-learning-potential thermodynamic verification → sol-gel synthesis → XRD/charge–discharge characterization), recommending NMC-Si(Mg/Ca/B) cathodes 18.5–28.8% above NMC811 [11].

Materials-science agents provide a transferable *propose–verify–feedback* closed-loop architecture for battery design: the LLM generates and justifies candidate designs; deterministic tools (DFT, machine-learning potentials, databases) verify them; and self-reflection or critiquer agents feed the results back into the next prompt. LLMatDesign’s modify–self-reflect loop converged on band-gap targets in 10.8 steps on average [12]; MASTER’s three-layer design–simulate–review multi-agent reached 97.2% geom-

etry success while saving up to 90% of atomistic simulations against single-agent baselines [13]; ChatMat’s manager–subagent division succeeded on 91–96% of four tasks, cutting average DFT calls from 1.15 to 0.35 [14]; MatClaw’s code-first pipeline [15] and MatSciAgent’s multi-task routing [16] belong to the same family. These systems also expose two recurring bottlenecks: implicit domain knowledge (simulation time scales, equilibration protocols, convergence criteria) still requires expert constraints or literature injection; and spatial/structural reasoning plus verification realism remain weak—AtomWorld reports success rates below 12% on rotation-type structural operations, positioning LLMs as co-pilots rather than full autonomous modelers [17], and AlchemyBench’s evaluation of synthesis-recipe extraction still leans on externally curated expert labels with automated extraction sensitive to source-document errors [18]. Most critically, a fully closed synthesis–characterization wet-loop exists only in ChatBattery, and its wet-lab data are not yet fed back into the model automatically (the follow-up Li-rich variants were proposed by the researchers) [11].

Battery design is thus transitioning from data-driven search toward an intelligent-agent paradigm of knowledge-driven, mechanism-constrained, simulation–experiment closed loops, but existing work leaves three clear gaps: the simulation loop and the wet-lab loop remain severed, experiment data not feeding back into model re-iteration; explicit physics constraints and interpretability in agent hypotheses are insufficient; and there is no standardized evaluation benchmark spanning the whole battery design process (inverse estimation, multi-objective trade-offs, long-duration reliability). The present work targets a dimension all three gaps share and none of the surveyed lines addresses: *who judges the design?* Both existing baselines either pin the measurement standard by construction (a black-box optimizer on a fixed objective) or leave

it free (an ungoverned LLM agent on a natural-language contract that it also reports against). Our study occupies the empty quadrant: a governed agent executing the full chain of cell-design levers (electrolyte, separator, electrode, architecture, thermal) in a single autonomous loop, with every verdict computed by an adjudication layer outside the agent. To our knowledge, no published system spans that chain within a single autonomous loop; in this system, judgment is computed by an adjudication layer outside the agent.

1.2. The Governance Question

Four properties of the system follow from that design. First, the contract is the only input: a natural-language requirement with nothing else (no configuration file, no parameter set, no candidate list) and the agent returns the complete, adjudicated design: material choices, formulation, architecture, safety validation, and the deliverable package, with a closing first-principles endorsement pass for the flagship molecular finalists (§5.8). Second, governance-carrier decoupling: the protocol is a self-contained declarative artifact (natural-language rules, a command-line tool library, and an audit ledger) bound to no specific agent framework, a property we verify empirically by running it under three orchestration substrates (§5.7). Third, the human engineer’s *in-loop* role ends at the moment the contract is stated. Fourth, the eight scenario contracts were chosen so that different tasks exercise different levers of the full chain (a fast-charge guarantee, an SEI-kinetics bridge for a lifetime bound, a cathode-system switch for a voltage plateau), which the governed agent exercises in turn, in the same loop, under the same protocol: the full-chain claim is suite-level, the suite collectively spanning all five design domains. This is a delegation setting no agent-based design system has yet been able to study: a complete design chain handed

to a machine, with the standard-setter present only as a written contract.

What makes such delegation safe (or dangerous) is the question this paper answers. We deliberately construct the two failure modes a design organization would most fear. The first is the *solution-space failure*: the strongest classical baseline, Bayesian optimization, moves only architecture parameters; when a contract requires a material lever (a high-voltage spinel chemistry, an SEI-suppressing coating) the optimum is not in its reachable set. The second is more subtle and, we will argue, more dangerous: the *yardstick failure*. A protocol-free agent with the same tools, budget, and model is not constrained to measure against the contract it was given, and in our data adjusts it in five distinguishable modes (Section 5.5)—a self-set threshold (“dendrite onset at -10 mV”), a substituted model, a purchased physical parameter, an altered caliber, and an omitted criterion variable. In our data both baselines fall short for orthogonal, specifiable reasons (the optimizer threefold, bounded by its space; the bare agent zero within the space despite claiming seven) one loses to an unreachable space, the other to a malleable measurement standard.

We introduce *protocolized governance* as the intervention: a fixed, always-on set of process rules the agent executes inside the design loop, plus an adjudication layer that stands outside it. Its four components: a layered multi-precision funnel, evaluation-and-fallback routing, heterogeneous model voting, and, the one the results hinge on, *mechanized adjudication*: are specified in Section 3. Under mechanized adjudication the contract is pre-registered as machine-readable criteria in an audit ledger before the first simulation runs; every verdict is computed by code from tool-output files, not asserted by the agent in prose; each “achieved” claim points to the exact file and key that supports it; and a verifier refuses incomplete audit trails. In this system

the verdicts are *computed* rather than *asserted*. In the language of delegation theory, it is a delegation contract with enforced fidelity—the alignment between the contract the principal believes it set and the behavior the agent actually executes [19, 20].

We evaluate the governance in a thirty-two-run matrix plus twelve ablation runs. Eight scenario tasks: a sedan’s energy-and-fast-charge contract, a drone’s weight-and-rate envelope, a smartphone’s volumetric ceiling, and five others—are written as natural-language requirements with absolute numeric criteria and no other declarations. Each task is given, under identical resources, to three decision-makers: the governed agent (two different LLMs, separately and jointly); a protocol-free agent on the same harness with the same toolset and budget; and a Bayesian optimizer configured to the standard of the strongest comparable published agent framework [5]. The results answer the question sharply. The governed agent achieves **all eight** contracts under the two DeepSeek-generation models, and four of eight under each of two foreign-vendor models (Section 5.4). The BO baseline achieves three—precisely the three tasks whose bottlenecks are structural rather than material. The protocol-free agent achieves none within the same measurement space: of its seven self-reported successes, two rest on a lithium-metal cell outside the validated parameter sets and are excluded from the in-space count, five dissolve under contract-caliber re-adjudication, and the one session that made no claim exhausted its budget.

Three further results speak to *why* governance works. First, ablation shows the mechanisms are not interchangeable: turning off the ceiling assessment that triggers material escalation costs two of three material-bottleneck tasks outright; turning off forced architecture exploration costs one of six; turning off the three-model vote changes nothing: an “unrealized premium”,

since in this suite the governed agent never exercised the molecular path, preferring cheaper routes. Second, the governance is model-robust: eight of eight under both a flagship and an economy LLM, the same contracts attained through different designs. Third, the simulation world is anchored to a physical artifact: the beginning-of-life 0.1C discharge curve of a production 21700 cell (LG M50T, [21, 22]) matches our calibrated parameter set [23] to within 0.45% capacity and 20.6 mV RMSE—a validation of the simulator’s calibration against a production cell, as distinct from any validation of the designs themselves.

The paper makes three contributions. (i) We present the first end-to-end agent for full-chain battery cell design: given nothing but a natural-language scenario contract, a single autonomous loop traverses the whole chain of battery design levers—material system, formulation, cell architecture, thermal management, safety, and aging—and returns a complete adjudicated design whose conclusion-level numbers pass a multi-level true-compute endorsement (ML-potential screening, two independent DFT codes, and a classical-force-field diffusion simulation). The system is delivered as open infrastructure: the protocol rule set, the simulation tool library, the adjudication and audit layer (pre-registered criteria, code-computed verdicts, evidence chains, audit-chain rejection), and every experiment artifact are released in a single open-source repository, demonstrated portable across three agent harnesses (Section 5.7), and every conclusion-grade number can be mechanically replayed from the audit ledgers. (ii) We distill the system’s steering logic into a transferable mechanism pattern for engineering agents: multi-resolution cost scheduling (cheap proxies first, true first principles last), symptom-to-scale fallback routing (failures diagnosed to the scale of their cause before retrying), a ceiling assessment that triggers material-level search only when

the current system provably cannot close the gap, a three-consecutive-failure stop rule that questions the assumptions rather than retrying blindly, and heterogeneous-model disagreement detection (reported as not exercised in this suite, with the null documented). The pattern is agent-agnostic: it migrates to any multi-scale, multi-fidelity design workflow. (iii) We decompose the novelty modes reachable by a governed design agent: retrieval, cross-domain selection, and structural invention. Under a prose prohibition the agent retrieves documented additive compounds—reliable (all real, cross-checked) but novel in neither mode (T9). Under criteria-bound forcing it leaves the pre-registered 82-compound additive set and selects a cataloged compound, bis(trifluoromethyl)sulfone (CAS 72971-96-1), whose battery-additive use is unattested in the searched scope—reliable (full mechanical chain), use-novel (searched scope), structure-cataloged (T10). Structural invention is the one mode not attained, and not attestable by model or simulator. Together the two conditions answer the menu-selection critique: retrieval is verified and reliable within the menu, and the innovation contract design can deliver is not new-structure discovery but verified cross-domain selection.

1.3. Contributions and Organization

The remainder of the paper is organized as follows. Section 2 reviews autonomy in scientific and engineering design and situates governance in the delegation literature. Section 3 specifies the protocol and its implementation. Section 4 defines the experimental design and the contract-verification machinery. Section 5 reports results; Section 6 discusses and calibrates; Section 7 concludes with implications for the design and management of agent-mediated decision systems.

2. Related Work

This paper sits at the intersection of three literatures: autonomous agents in scientific and engineering design, classical optimization baselines and their known boundaries, and the governance of delegated decision-making. We review each in turn and then state the gap.

2.1. *Autonomous Agents in Scientific and Engineering Design*

The past three years have seen LLM agents deployed across the design chain, one slice at a time. Coscientist automated end-to-end chemistry workflows (planning, coding, instrument control) in a single agent [24]; ChemCrow augmented an LLM with chemistry tools for molecule design and synthesis planning [25]; the A-Lab closed the loop from synthesis recipe to robotic execution to X-ray characterization for materials discovery [26]. In the battery domain specifically, the lineage is young but active: LLM-assisted battery research has been surveyed comprehensively [8]; foundation models target battery analysis [9]; and multi-agent chemist systems extend the approach to material prediction and exploration [14, 16, 15], and further agent families span hierarchical multi-agent materials discovery [13], drug piloting and RAG-enhanced collaborative drug discovery [27, 28], and AMS circuit-design optimization [29]. Closest to our setting, Battery-Sim-Agent performs *inverse* battery parameter estimation with an LLM agent in a simulator-in-the-loop configuration, reporting 67–95% curve-matching error reduction against classical black-box optimizers across 200 synthetic calibration tasks [5]. Adjacent engineering domains replicate the pattern: AnalogAgent automates analog circuit design [30]; GENIUS orchestrates autonomous design and execution of simulation protocols [31].

Two structural observations hold across this entire body of work. First, every system covers a slice of a design chain, not a chain: material discovery stops at the material, circuit automation stops at the circuit, inverse estimation stops at parameters. None spans a full set of coupled design levers (from chemistry through formulation through architecture) and none therefore faces the cross-scale diagnosis problem that a full chain poses. Second, the judgment in each system is asserted by the agent itself. Whether a synthesis “worked” or a parameter fit “converged” is reported in natural language; there is no pre-registered contract against which a code-level verifier judges the claim. Benchmark evaluations of these systems reflect the same design: DiscoveryBench, ScienceAgentBench, MAgentBench, and RE-Bench measure task success and capability [32, 33, 34, 35]; FIRE-Bench adds an honesty layer for scientific rediscovery claims [36]. None treats the *integrity of the yardstick* (whether the metric an agent reports against is the metric it was given) as a first-order experimental variable.

2.2. Classical Optimization Baselines and Their Known Boundaries

Bayesian optimization (BO) is the standard strong baseline against which LLM agent systems are measured, and the comparison record already contains an important calibration. Battery-Sim-Agent benchmarks its agent against Meta’s Ax BO platform (Sobol initialization, GPEI with LogNEI acquisition, fixed seed) and reports not only the headline reduction but a calibrated boundary: in “low-headroom” regimes where literature default parameters already sit close to the target, BO remains competitive and can even outperform the agent, and CMA-ES fails to converge on their parameter estimation tasks altogether [5]. We adopt the identical BO configuration (same platform, same strategy family, same seed) so that our baseline inher-

its this published calibration.

The deeper boundary, which no agent study has isolated experimentally, is structural rather than statistical: BO searches a fixed parameter space. A classical optimizer can move thickness and porosity, but it cannot invent an electrolyte formulation, propose a coating, or switch a cathode system. When the binding constraint on a design contract is a material property—a voltage plateau beyond the chemistry’s polarization limit, an SEI growth rate that only a coating suppresses: the optimum is not in the reachable set, and no acquisition function can find it. The battery modeling literature supplies the physics that makes this boundary concrete: plating onset is governed by the local overpotential balance [37, 38]; SEI growth by reaction kinetics [39]; thermal runaway by the coupled decomposition chemistry [40]. Each of these mechanisms admits levers at scales a structural optimizer cannot touch. Our experimental design makes this boundary an observable: three of our eight contracts are solvable within the structural space, and five are not.

2.3. Agent Reliability and the Case for External Verification

The central obstacle to trusting an agentic design system is not model capability but measurement: an agent that both executes and evaluates can move the yardstick it is measured against. The AI community has begun to quantify this. Honesty benchmarks establish that LLMs systematically overstate what they know [41, 42]; FIRE-Bench shows that even frontier scientific agents fail most reported reconstructions, and that their self-reported completion is not a reliable signal of actual convergence [36]. Evaluation practice has responded with two remedies. The first is *soft judging*: LLM-as-judge and reward-model reviewers applied to agent outputs. The second is *hard*

judging: deterministic, code-based evaluation of claims against ground truth (as in unit-test-style benchmark harnesses). Hard judging is the technically reliable option, but in existing systems it evaluates **the benchmark**, not the agent’s own workflow; the design loop itself is never forced under a deterministic evaluator. The management literature arrives at the same conclusion from the opposite direction: delegated agency theory [19] identifies contract specification quality and governance visibility as the design dimensions that discipline delegated action; empirical work on AI-assisted decisions shows algorithmic recommendations reshape human override behavior [20]; Bayesian delegation policies allocate decision authority sequentially [43]; GAIA articulates information-gated progression for high-stakes LLM agents [44]; and qualitative work warns that autonomy must be designed into systems rather than assumed [45]. What none of this literature contains is an *engineering-level* instantiation: a concrete design environment in which the evaluator is code, the evaluation contract is pre-registered before work begins, and the fidelity of that evaluation is itself an experimental variable.

Finally, we note the disciplinary bridge this paper crosses. The battery prognostics community rooted in grey system theory (originating in the management science tradition [46]) has long modeled battery degradation and state-of-charge as forecasting problems [47, 48, 49, 50, 51, 52]. That lineage treats the battery as an object to be *predicted*. This paper treats it as an object to be *designed by a governed agent*, extending the same management-science lens from prognostics to design governance.

2.4. The Gap and Research Questions

Three gaps follow. (1) No published system spans the full chain of cell design levers (electrolyte, separator, electrode, architecture, thermal) in one

autonomous loop. (2) No experimental comparison isolates the two failure modes of ungoverned delegation: the solution-space failure of classical optimizers and the yardstick failure of protocol-free agents. (3) No study treats the adjudication layer (pre-registered contracts, code-mechanical verdicts, evidence-chain audits) as an experimental variable whose removal can be measured. The paper is designed to isolate both failure modes: the solution-space failure and the yardstick failure.

We formalize three research questions:

- RQ1** *Governance performance.* Does protocolized governance raise contract attainment relative to a protocol-free agent and a Bayesian optimizer under identical resources, and at what resource (token and message) cost?
- RQ2** *Mechanism attribution.* Which governance components carry the effect, and in which task regimes? We ablate each component separately and report its marginal contribution, including null results.
- RQ3** *Yardstick trustworthiness.* Does mechanized adjudication close the judgment-integrity gap (measured as evidence completeness, criterion fidelity, and the absence of self-set thresholds) that the protocol-free agent exhibits?

3. System Framework: Protocolized Governance

3.1. The Design Space and the Contract

A battery cell design is one indivisible decision spanning five interlocking design domains: electrode chemistry, electrolyte formulation and transport

matching, electrode microstructure construction and modification, architecture parameters (thickness, porosity, separator, current collectors, particle radius), and cooling topology, each layer constraining the next, such that an optimum at any single domain can induce system-level performance regression or physical infeasibility. For decades the only infrastructure capable of making this decision end-to-end was a multidisciplinary human team, together with its coordination costs: communication loss, version conflicts, knowledge silos. Automation has stopped at the boundary: Bayesian optimization adjusts architecture knobs inside a single chemistry [5], autonomous laboratories close the synthesis loop [26], LLM chemists execute molecular tasks [24, 25]. Each covers one segment; none has spanned the full chain (electrolyte to thermal integration) within a single autonomous closed loop.

The system we describe does. In the virtual battery factory, a single LLM agent receives a natural-language requirement and returns a complete adjudicated cell design covering materials, formulation, architecture, safety validation, and a first-principles endorsement pass. The system’s core design principle is *governance-carrier decoupling*: the protocol exists as a declarative natural-language rule set bound to no specific agent framework and portable across orchestration environments (demonstrated empirically in Section 5.7). The human engineer’s role ends at the moment the contract is stated; the agent then plans, screens, simulates, diagnoses, falls back, and concludes with no intermediate human oversight. The eight scenario contracts were chosen so that each exercises different levers of this chain, and the suite collectively spans all five design domains.

The design space a battery cell exposes is organized into five lever families, each with a complete simulation chain: the *electrode system* (switching among validated parameter sets: NMC811/graphite-SiOx [1, 23], the

degradation-coupled NMC set [53], LFP [54]); the *electrolyte formulation* (transport properties σ , t^+ , D_e with their temperature dependence [55], solvent/salt formulation, additive packages); the *electrode modification* (coatings and dopants acting on SEI kinetics [39] or cracking rates); the *cell architecture* (electrode thickness, porosity, N/P ratio, separator, current collectors, particle radii); and *thermal management* (the cooling coefficient). A sixth family (solid-phase conductivity and diffusivity) is deliberately excluded as a design lever because no formulation model maps them, so any adjustment would be an unverifiable estimate: the protocol treats “what can be simulated” as the boundary of “what may be designed”, and estimates may not masquerade as simulation output. A direct consequence is the material-innovation frontier of this experiment: a novel additive or composition has no simulator parameter counterpart without kinetic-level data, so material design in the operational sense runs through system switches, electrode-modification bridges, formulation matching, and screening of literature candidates.

The contract is the only input. A task is a natural-language requirement with absolute numeric criteria—“design a battery for a sedan: energy density ≥ 392.61 Wh/kg, 4C fast charge without lithium plating, maximum temperature $\leq 60^\circ\text{C}$, overcharge to 4.7 V without thermal runaway”, and nothing else. There is no configuration file, no parameter set assignment, no candidate list, no round budget: every case-level decision (which parameter set, which starting stage, which levers are adjustable, which safety defaults apply when the text is silent) is derived by protocol rules from the text alone, and the derivation is written into the audit ledger before the first simulation. The task text’s numbers are the contract verbatim; the agent may not re-anchor, re-calibrate, or re-interpret them. Figure 2 summarizes the

system: the contract enters the governed loop, and the adjudication layer writes every verdict into the audit ledger outside it.

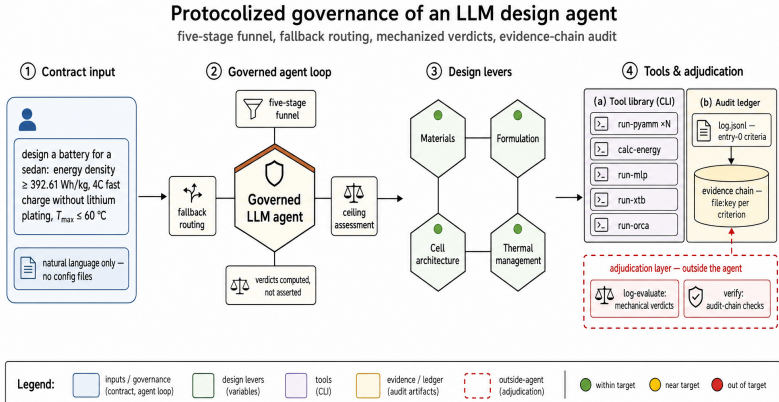


Figure 2: System architecture. The natural-language contract is the sole input; the agent executes the five-stage loop over a command-line tool library; the adjudication layer sits outside the agent and writes mechanized verdicts with per-criterion evidence into the audit ledger.

3.2. The Pipeline and the Five Governance Components

The protocol organizes work as a five-stage funnel: planning, molecular screening, cell design, safety assessment, and a closing true-compute endorsement—inside which four protocol components operate as fixed, always-on rules. Figure 3 shows the loop with its fallback routing and the outside-agent adjudication layer.

Component 1: the layered multi-precision funnel with ceiling escalation. Screening proceeds from cheapest to dearest: millisecond-scale machine-learning potentials (MACE [56], CHGNet [57]) and a semi-empirical quantum method (xTB [58]) at the molecular scale; seconds-scale SPM_e then minutes-scale DFN at the cell scale. The opening of cell design triggers a

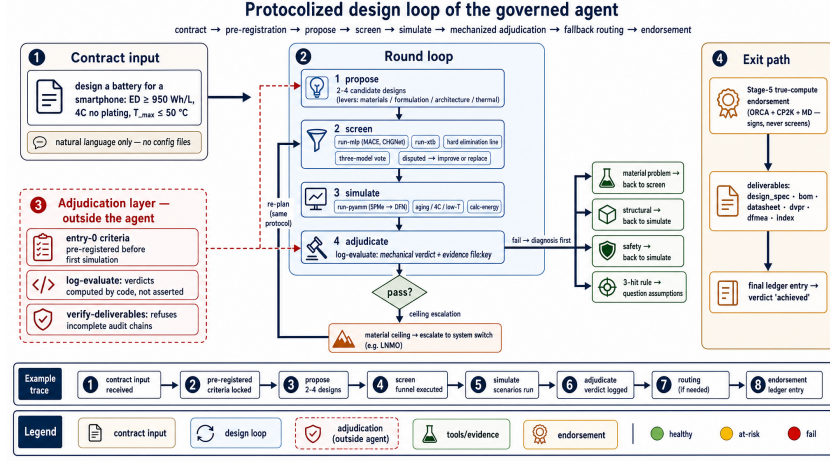


Figure 3: The protocolized design loop. The contract is the sole input; each round proposes, screens (when the bottleneck is molecular), simulates, and is adjudicated by code outside the agent; diagnosed failures route back to the scale of their cause, and a material ceiling escalates to a system switch. The red-dashed adjudication layer delivers the mechanical verdict and refuses incomplete audit chains; the exit path ends in endorsement and the deliverable package.

ceiling assessment: the agent estimates the best architecture-plus- formulation achievable within the current system and compares it to the contract. If the objective exceeds the ceiling, the agent is required to escalate into material design (inventing additives or switching systems) rather than tuning parameters inside a space that provably cannot close the gap:

$$\text{escalate} \iff \exists c : t_c > \text{ceiling}_c(\text{system}), \quad (1)$$

where $\text{ceiling}_c(\text{system})$ is the best contract value the current system’s architecture and formulation can reach on criterion c . The escalation decision is written to the log with its evidence (which levers were assessed, the limit values, the gap).

Component 2: evaluation and fallback routing. Every stage out-

put is evaluated against the contract immediately; on failure, the agent must diagnose the scale at which the cause lives and fall back there (material problems to the molecular stage, architecture problems to cell design) not blindly rerun the pipeline. A three-consecutive-failure discipline is attached to the fallback: three consecutive failures with the same cause trigger a mandatory re-examination of the assumptions (“is this objective physically reachable in this system?”) rather than a fourth blind attempt. The symptom-to-scale correspondence is fixed: potential windows and stability are molecular properties; capacity and temperature rise are architecture properties.

Component 3: heterogeneous model voting. Every molecular candidate is screened by three methodologically distinct models (two ML potentials from different families plus a semi-empirical quantum method) whose agreement or disagreement is itself signal. Hard elimination lines (energy above the stability ceiling, non-convergence, HOMO above the oxidation line) are mechanical; disagreement among the three ranks marks a candidate *disputed*:

$$\text{disputed} \iff \text{spread}(r_1, r_2, r_3) \geq \max(2, 0.3N), \quad (2)$$

where r_1, r_2, r_3 are the candidate’s ranks under the three models among N candidates—forcing reasoning-based refinement rather than acceptance or rejection on a single model’s word.

Component 4: mechanized adjudication and the audit chain.

Unlike the other components, this one pins the outcome itself. Before the first simulation, the agent writes entry 0 of the audit ledger: the contract criteria, pre-registered in machine-readable form and layered by decision stage, with case-level metadata recording every rule-derived decision and its basis. Thereafter the agent is *forbidden from judging*: every evaluation must

be issued by the `log-evaluate` command, which reads the criteria and the tool-output files, compares each metric against its threshold in code, and emits a verdict with an evidence chain: for each checked criterion, the file, the key, the value, and the threshold it was compared against. A design is “achieved” exactly when

$$\text{achieved} = \bigwedge_c (v_c \geq t_c^{\min} \wedge v_c \leq t_c^{\max}), \quad (3)$$

where the conjunction ranges over the pre-registered criteria of the contract. Derived criteria are computed mechanically; the plating criterion is the minimum of the anode potential time series against zero, never a threshold the agent may pick:

$$\text{plated} = \left[\min_t \phi_{\text{anode}}(t) < 0 \right]. \quad (4)$$

Equations (3)–(4), like every criterion expression in this section, are evaluated verbatim by the adjudication layer. A closing verifier (`verify-deliverables`) checks the deliverable set and, crucially, the audit chain itself: every proposal round must have a same-round mechanical evaluation, and a final judgment entry must exist—a run whose ledger lacks either is refused as incomplete. Judgment is thus an infrastructure layer outside the agent, following the separation of duties in auditing: the entity under audit does not write its own verdicts.

The protocol, the tool library, the adjudication layer, and the cross-harness harnesses are open source (github.com/wangyuanxty/BatteryDesignAgent).

The fifth pipeline stage is the true-compute endorsement: for the closing top candidates only, ab initio DFT (ORCA, r2SCAN-3c [59]) with optional cross-validation in an independent code (CP2K, PBE-D3 [60]), and molecular dynamics for the top electrolyte diffusion coefficient. The endorsement *signs* conclusions; it never screens. In the experiment matrix reported here the

endorsement runs are executed for the flagship cases and reported separately (Section 5.8); turning it on or off does not change any contract verdict, by design.

3.3. The Protocol as a Declarative Artifact

The protocol is packaged as a self-contained declarative artifact: a natural-language rule set, a command-line tool library (twenty-plus simulation, evaluation, and audit commands), and the audit ledger format; in the open-source release it is distributed as the project’s skill package (SKILL.md together with the reference documentation and the tool library), so the rule set and the tools an agent executes against are versioned and installed as one unit. It binds to no agent framework—any harness that can loop a model over a shell and files can execute it. The governance travels with the contract, not with the harness. In the present study every run executes under one harness; cross-harness portability is a structural property by construction, and its empirical demonstration (the same protocol under additional agent frameworks) is demonstrated empirically with identical contract outcomes (Section 5.7).

3.4. Implementation

The tool library is implemented as a Python package over PyBaMM [4] for electrochemistry—including the thermal-electrochemical parameters [23] and degradation physics [53, 39, 37, 38, 40]: plus surrogate and true-compute runners. Twenty-plus commands cover simulation protocols (discharge at 1C/0.1C/5C, 4C charge at 45 °C, low-temperature discharge, overcharge, 100-cycle aging, nail penetration with a three-reaction thermal-runaway ODE), contract-caliber energy accounting, the mechanical evaluator, and the deliverable verifier. The adjudication layer is pinned by an automated regression

suite distributed with the tool library. All conclusion-grade numbers in every report originate in tool outputs; the ledger is the paper’s primary data source, and every figure and table here is derivable from it.

4. Experimental Design

4.1. Task Suite

The task suite consists of eight scenario tasks, each a natural-language requirement written as an application scenario plus absolute numeric criteria (Table 1). The scenarios span the design landscape deliberately: a sedan (performance plus thermal safety), grid storage (cycle life plus low temperature), power tools (rate plus power), extreme cold (low temperature plus volumetric density), a flagship vehicle (extreme energy density), a smartphone (volumetric ceiling plus a strict temperature red line plus a voltage plateau), a hybrid vehicle (high-temperature aging plus nail penetration), and a drone (specific energy plus a mass ceiling). Across the eight tasks, fourteen distinct metric types are adjudicated, and all criteria of a task are AND-judged: a design must satisfy every clause of its contract to be “achieved.”

Three design choices deserve emphasis. First, the tasks carry no constraint declarations beyond the metrics: nothing states which levers are adjustable, which chemistry to use, or where the baseline sits. Undeclared items are interpreted in the widest sense under protocol rules, with the derivation recorded—this is what forces the agent to make the design-space decisions itself. Second, every number in every task is absolute; no relative-baseline phrasing appears, so an agent cannot redefine the reference it measures against. Third, the tasks are heterogeneous in where their binding constraint

lives: T1, T4, and T8 are solvable inside a structural space (thickness, porosity, mass), while T2, T3, T5, T6, and T7 require material, formulation, or transport levers—the distinction on which the comparison with the Bayesian baseline turns.

The contract quantities are defined on standard cell models from the cited literature. Plating is judged on the negative-electrode potential against Li/Li^+ , written as the sum of the equilibrium potential and the kinetic overpotential [37, 38]:

$$\phi_a = U_a + \eta_c. \quad (5)$$

The SEI criterion reads a state variable of the degradation model, whose growth law supplies the coating lever that T2 and T6 exploit [39, 53]:

$$\frac{d\delta}{dt} = k_{\text{SEI}}, \quad (6)$$

where k_{SEI} is exactly the parameter the electrode-modification bridge adjusts. The thermal-runaway trigger for the overcharge and nail-penetration criteria is the three-side-reaction energy balance [40]:

$$m c_p \frac{dT}{dt} = \sum_k \Delta H_k r_k(T) - h A (T - T_{\text{amb}}), \quad (7)$$

with triggered iff the trajectory crosses the runaway onset; the cooling coefficient h is the thermal-management lever. Criterion values are model-relative (an SEI limit reads the state variable of a specific degradation model, a plateau reads a specific OCP set) so cross-system comparisons are made only within the same model. The scenario anchoring (which production specification each task draws from, and how each threshold was calibrated from the underlying parameter sets) is documented in the supplementary materials; each threshold is the anchoring parameter set’s baseline value for that

criterion shifted by a fixed increment (e.g., +20% energy density over the baseline configuration, a 5-point cut on 5C retention), never a frontier-tight value—the experiment measures governance contribution at fixed contracts, not frontier performance. The anchoring is a Methods-section explanation and never appears in the task text seen by any agent.

4.2. Conditions and Control Groups

The design follows a treatment-plus-two-controls structure. The governed agent is the treatment; two control groups isolate, respectively, the protocol itself and the classical-optimization boundary. All three conditions share the task texts, the simulation stack, and the session budget, so each observed difference is attributable to the single variable that condition removes.

Governed agent (G). The full protocol of Section 3, executed under two large language models separately (deepseek-v4-pro and deepseek-v4-flash) giving eight runs per model and a model-robustness axis. Two further legs re-execute the eight sessions under GPT-5.6 Luna and under mimo-v2.5 (foreign-vendor models served through OpenAI-compatible APIs, same protocol, harness, task texts, and budget) so the cross-family comparison is made outside the primary vendor at matched tier logic.

Protocol-free agent (C1). The same LLM, the same agent harness (tool-use loop), the same tools available in the environment (a Python interpreter with the simulation libraries, shell, and file tools), and the same session budget—but no protocol: the system prompt carries only a role statement, the tool descriptions, a ban on true-compute calls, the budget, and the deliverable request. There are no stage rules, no funnel, no pre-registered criteria, no mechanical evaluator, and no audit-chain verifier. C1 is the

Table 1: The eight scenario tasks. Every criterion is an absolute number, AND-judged; nothing else is declared in the task text.

ID	Contract (abridged; all values absolute)
T1	Sedan: energy density ≥ 392.61 Wh/kg; 4C charge without lithium plating; $T_{\max} \leq 60^\circ\text{C}$; overcharge to 4.7 V without thermal runaway.
T2	Grid storage: energy density ≥ 327.18 Wh/kg; 4C charge without plating; SEI ≤ 500 nm after 100 cycles; $\geq 90\%$ capacity retention at -20°C ; SEI ≤ 550 nm after 500 cycles.
T3	Power tools: capacity ≥ 2 Ah; 5C retention $\geq 95\%$; 4C charge without plating; $T_{\max} \leq 60^\circ\text{C}$; power density ≥ 4000 W/kg.
T4	Extreme cold: $\geq 95\%$ 1C retention at -20°C ; energy density ≥ 327.18 Wh/kg; volumetric energy density ≥ 880 Wh/L.
T5	Flagship: energy density ≥ 500.94 Wh/kg; 4C charge without plating; $T_{\max} \leq 60^\circ\text{C}$.
T6	Smartphone: volumetric energy density ≥ 950 Wh/L; 4C charge without plating; $T_{\max} \leq 50^\circ\text{C}$; SEI ≤ 500 nm after 100 cycles; voltage plateau ≥ 4.1 V.
T7	Hybrid: energy density ≥ 327.18 Wh/kg; 4C charge without plating; SEI ≤ 550 nm after 100 cycles at 45°C ; nail penetration (10 W short-circuit heat) without thermal runaway.
T8	Drone: energy density ≥ 446.18 Wh/kg; 5C retention $\geq 90\%$; cell mass ≤ 40 g.

protocol-removal control: it isolates the governance layer by holding every other resource constant: model, harness, toolset, budget, task texts. One

run per task, deepseek-v4-pro.

Bayesian optimization (C2). A classical BO baseline in the architecture parameter space (ten structural parameters: electrode thicknesses and porosities, separator thickness and porosity, current collector thicknesses, heat transfer coefficient, negative particle radius), searching the objective *energy density minus safety penalties*,

$$\text{obj} = \text{ED} - \sum_k \lambda_k \mathbb{K}[v_k \not\geq t_k], \quad (8)$$

with a fixed penalty set that covers every contract criterion except the T1 overcharge-runaway trigger (plating, temperature limit, SEI, plateau, mass); a criterion absent from this set is invisible to the search. The material-bottleneck failures are therefore not a visibility artifact: SEI and plateau *are* penalized (a re-run with the aligned 50 °C red line confirms the failures persist, Section 5.1). What the optimizer cannot do is move a chemistry switch, because its knob space is structural only—and that boundary is the point: the configuration replicates Battery-Sim-Agent’s published BO standard [5]: Meta’s Ax platform, Sobol initialization, GPEI with LogNEI acquisition, fixed seed 1234, fifty evaluations per task—the same simulator and the same contract-caliber energy accounting as the agents, so the comparison is on identical physics. The search is deterministic in its seed; a three-seed replication (1234, 5678, 9012) reproduces the attained-task set $\{T1, T4, T8\}$ identically, with per-task best objectives within 9% across seeds (supplementary materials). C2 is the *classical-baseline control*: it isolates the solution-space boundary of ungoverned search: a strong, published-standard optimizer that cannot leave its parameter space. One run (fifty evaluations) per task.

Resources held constant. All three conditions receive the same task texts, the same simulation stack, and the same session budget: the agents

run under the same harness turn budget (the agent SDK’s `max_turns=300` setting) and the protocol carries no round cap of its own. Table 4 reports a log-derived activity metric, *assistant messages*, which is finer-grained than the harness’s internal turn counter; the budget *setting*, not the message count, is the controlled resource. The two units are empirically anchored: the one session that hit the budget (the protocol-free agent on T6) did so at 443 assistant messages. Every other session concluded before the cap. The BO baseline’s fifty evaluations define its budget. A second, simulator-native currency is likewise anchored: the governed sessions average 25 run-pyamm calls per task (range 0–55; 31 including calc-energy), against BO’s fixed fifty: the agents hold no evaluation-count advantage, and both conditions sit in the same order of magnitude on every resource axis. Neither currency is equalized per operation; each condition receives its method family’s standard budget, and the comparison is made at that level. True-compute endorsement is disabled across the main matrix (`real_compute = false`); endorsement results are reported separately in Section 5.8.

4.3. Ablation Design

The governance components are fixed properties of the protocol; ablation is an experimental intervention that relaxes exactly one component per run. An ablation variant is declared in the task text itself (“ceiling escalation disabled”) (keeping the natural-language requirement the sole entry point) and the declaration is recorded in the run’s entry-0 metadata. Three ablations are executed: *ceiling escalation off* (no proactive material-design escalation; the agent tunes within the current system unless the text names materials), on T2/T5/T6; *architecture forcing off* (no required 2–4 architecture variants per round), on T1/T2/T3/T4/T7/T8; and *funnel voting off*

(molecular screening by a single ML potential instead of three-model heterogeneous voting), on T2/T5/T6. Each ablation run uses deepseek-v4-pro and the full protocol otherwise. Twelve ablation runs in total.

4.4. Judgment Machinery

All verdicts reported in this paper come from the mechanized adjudication layer of Section 3: criteria pre-registered in entry 0, verdicts computed by `log-evaluate` from tool-output files, evidence chains per criterion, and the deliverable verifier’s audit-chain checks. Two further audits are applied to the control conditions. For C1, the agent’s own “achieved” claims are re-adjudicated at the protocol’s contract caliber by the authors: every claimed metric is recomputed mechanically from the artifact files the agent produced, on the protocol’s own definitions. Two calibers recur throughout and are fixed here:

$$\text{ED} = \frac{\int V(t) I dt}{\sum_i l_i (1 - \varepsilon_i) \rho_i \cdot A}, \quad (9)$$

the contract-caliber energy density (mass = layer stack, electrolyte excluded), and

$$\text{ret}_{5C} = C_{5C}/C_{1C}, \quad (10)$$

the rate-retention caliber against the 1C reference. The power criterion is computed from the discharge-start voltage drop and the cell mass (standard definitions of DC resistance and peak power):

$$\text{DCR} = \frac{V(0) - V(t_{10\%})}{I}, \quad P_{\max} = \frac{V_{\text{oc}}^2}{4 \text{DCR} \cdot m}, \quad (11)$$

with $V(t_{10\%})$ the voltage at 10% of discharge time. A claim computed against a different reference (e.g., 5C over C/3) or a different mass model is a different caliber and is recomputed (nail penetration means a thermal-runaway trigger

criterion, not a steady-state temperature). For C2, the objective trajectory is recomputed from the logged evaluation outputs. The re-adjudication rules are mechanical and are listed in the supplementary materials.

4.5. *Real-Data Anchor*

To anchor the simulation world to a physical artifact, the calibrated parameter set for the LG M50 cell [23] is compared against the beginning-of-life 0.1C discharge of a production LG M50T 21700 cell from the Imperial College degradation dataset [21, 22]: measured capacity 4878.9 mAh versus simulated 4857.0 mAh (+0.45%), and voltage-curve RMSE 20.6 mV over 2–82% depth of discharge. The alignment is the Methods-section calibration record; it establishes that the numbers agents are judged against are anchored to a production cell: a validation of the simulator, not of the final designs (Section 6). Figure 4 overlays the measured and simulated curves.

4.6. *Cross-Harness Portability*

The protocol’s independence from any specific agent framework (Section 3.4) is a design property of the artifact itself; its empirical demonstration is reported here and executed in Section 5.7, so that its results slot into Section 5 without methodological ambiguity. The design: the identical protocol (the same declarative rule set, the same command-line tool library, and therefore the same adjudication layer) runs the same task texts (T1, a structural-bottleneck contract, and T6, a material-bottleneck contract) under the same LLM and the same nominal budget setting (300, in each harness’s own turn unit) across three harnesses: the harness used throughout this paper, the OpenAI Agents SDK, and LangChain. The measured quantities are contract attainment (mechanically judged, identical criteria) and

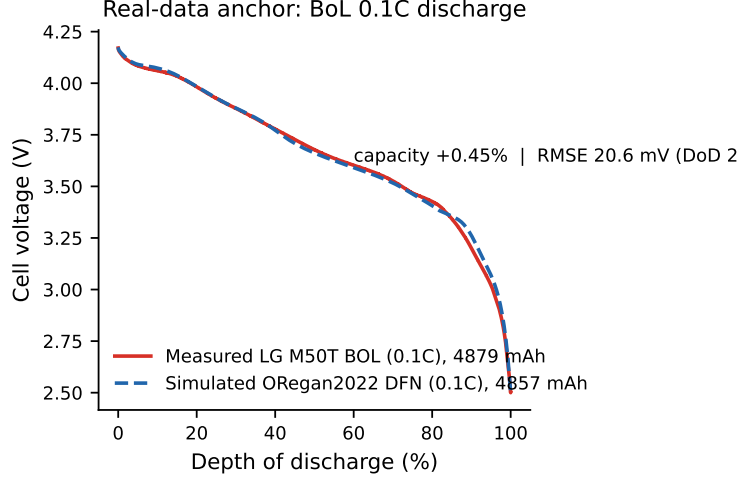


Figure 4: Real-data anchor: beginning-of-life 0.1C discharge of a production LG M50T 21700 cell (Imperial College dataset, [21]) versus the ORegan2022-parameterized DFN simulation [23]. Capacity agrees to +0.45%; voltage RMSE 20.6 mV over 2–82% depth of discharge.

audit-chain completeness (the verifier’s checks). Because the adjudication layer is a command-line tool rather than harness code, the verdict mechanics are harness-independent by design; the experiment measures whether agent behavior under the same governance remains contract-faithful when the orchestration substrate changes, as reported in Section 5.7.

5. Results

5.1. Main Comparison: Contract Attainment

Table 2 reports contract attainment for all eight tasks under the six columns. Three findings stand out. First, the governed agent achieves **8/8** under the two primary models and **4/8** under each of the two foreign-vendor models (Section 5.4); every verdict is mechanical, with an evidence chain

per criterion and a verifier-passed audit chain. Second, the Bayesian optimizer achieves **3/8**: exactly T1, T4, and T8, the three tasks whose binding constraint is structural (thickness, porosity, mass). Its best designs reach energy densities above the governed agent’s on these tasks (e.g., 678 versus 472 Wh/kg on T4), the expected consequence of an unconstrained thin-electrode search against a margin-balanced design; on the remaining five tasks it fails on the contract’s own criteria: 4C plating in T2, T3, T5, T6, T7; SEI limits in T2, T6, T7; a catastrophic 3.44% 5C retention on T3; the 4.1 V plateau unreachable on T6, where the best design stalls at the NMC811 plateau limit (4.0065 V)—a re-run with the penalizer aligned to the contract’s 50 °C red line (the main run hardcoded 60 °C) leaves the outcome unchanged: 0/50 trials attain the contract in either configuration, 30/50 clear the volumetric criterion alone in the aligned run, and none is simultaneously plating-free and below the red line; nail penetration triggered on T7. Figure 5 shows the BO trajectories: the three structural tasks climb to attainment, the five material-bottleneck tasks plateau against penalties no structural parameter can remove. Third, the protocol-free agent achieves **none** within the measurement space: two of its claimed successes (T4, T8) solve the contract by switching to a lithium-metal cell outside the validated parameter sets, and its remaining claims fail mechanical re-adjudication (Section 5.5). The tie between the two baselines (three achieved tasks each) with orthogonal failure modes is the paper’s central observation: the bare agent and the black-box optimizer fail for different, specifiable reasons, and only the governed agent passes on all eight contracts.

Contract-margin calibration deserves explicit reporting, since “8/8” could otherwise read as loose thresholds. The thresholds are fixed increments over the anchoring parameter sets’ baseline values, not frontier-tight values (Sec-

tion 4.1), and all three conditions face identical contracts: any looseness benefits the baselines equally. The achieved margins show the contracts binding where they are meant to bind: on six of the eight governed runs the last criterion to clear sits within 4.5% of its threshold (T6’s plateau at +0.4% and thermal line at +0.5%; T8’s mass at +0.5%; T3’s and T5’s thermal lines at +1.0–1.5%); the two exceptions are T2’s low-temperature retention (+10.6%) and T7’s SEI limit (+15.4%). The wide margins concentrate on the energy-density clauses of the structural tasks (+41–48%), where an unconstrained thin-electrode search overshoots cheaply; the same mechanism behind the BO’s 678 versus 472 Wh/kg on T4, and precisely the margin the governed designs partially spend on the safety and durability criteria. Per-criterion margins for all achieved runs are tabulated in the supplementary materials. The SEI criteria deserve one further disclosure, because they are the single criterion class whose bridge lever (the electrode-modification coating scaling the SEI kinetic rate constant) can move a verdict: T2 shipped a $10^{-4}\times$ rate constant (declared in the audit as an estimate beyond the bare-ALD literature magnitude), yet the identical design passes the contract at $10^{-2}\times$ (500.4 nm against the 550 nm limit, a 9% margin); T6’s SEI work is not verdict-bearing; T7 used no kinetic override (supplementary materials).

The thresholds are author-anchored, and one threshold property should be stated in advance of the reader’s own check: the comparison’s direction is invariant under a uniform perturbation of them. On the five material-bottleneck tasks the optimizer’s failure is ceiling-bound, not threshold-bound—across all 400 baseline trials, the highest midpoint voltage any design reaches is 4.0115 V against the 4.1 V contract line, and plating and SEI violations are lever-bound (supplementary materials). On the three structural tasks the same criteria bind both conditions in parallel: under a uniform +5% tighten-

ing the optimizer’s attained counts fall from 25/34/10 to 0/22/1, while the governed runs’ own binding margins (+2.2%/+4.5%/+0.5% on T1/T4/T8, Appendix B of the supplementary materials) fail under the same tightening on the same tolerance. No $\pm 5\%$ perturbation inverts the direction of the comparison; the only conceivable inversion zone—energy-density thresholds above the governed designs’ own margins, $> +40\%$ on the structural tasks—is exactly the region in which the paper already reports the optimizer’s structural-task energy-density superiority.

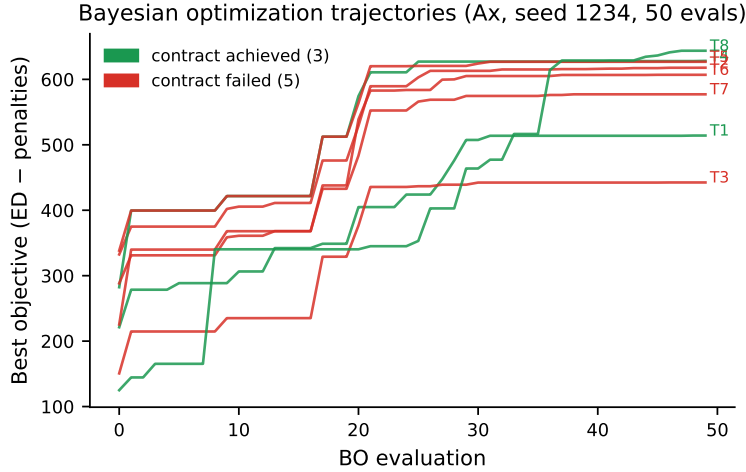


Figure 5: Bayesian optimization best-objective trajectories per task (Ax, Sobol initialization, GPEI + LogNEI, seed 1234, 50 evaluations). The three structural tasks (T1/T4/T8) achieve their contracts; the five material-bottleneck tasks plateau against penalties (plating, SEI, plateau, nail) that no structural parameter can remove.

5.2. Resource Cost

Table 3 reports the governed agent’s resource consumption per task under both models: assistant messages (activity measure) and model tokens (input/output) and Figure 6 plots the same data, highlighting the material-

Table 2: Contract attainment, eight tasks $\times$ six columns (governed agent shown under four models: pro, flash, luna, mimo). $\checkmark$ = achieved (mechanical verdict); $\times$ = failed contract criteria; $\circ$ = outside the measurement space (see text). G = governed agent; C1 = protocol-free agent; C2 = Bayesian optimization. Key re-adjudicated numbers in parentheses. Per-task numbers are r1 values; the condition totals are seed-robust as detailed in the table note and supplementary materials.

Task	G (pro)	G (flash)	G (luna)	G (mimo)	C1	C2
T1 sedan	$\checkmark$ 553.98	$\checkmark$ 459.4	$\times$ (plating)	$\times$ (plating)	$\times$ (−6.9 mV)	$\checkmark$ 514.13
T2 storage	$\checkmark$ 465.62	$\checkmark$	$\checkmark$	$\checkmark$	$\times$ (plating, SEI)	$\times$ (plating, SEI)
T3 tools	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$ (5C 84.8%)	$\times$ (plating, 5C)
T4 cold	$\checkmark$ 471.55	$\checkmark$	$\checkmark$	$\times$ (plating)	$\circ$ (Li-metal)	$\checkmark$ 678.26
T5 flagship	$\checkmark$	$\checkmark$	$\times$ (ceiling)	$\checkmark$ 517.8	$\times$ (−12.6 mV)	$\times$ (plating)
T6 phone	$\checkmark$ 1135.8	$\checkmark$	$\times$ (895.6 Wh/L)	$\times$ (852.8 Wh/L)	$\times$ budget	$\times$ (plateau, $T_{\max}$)
T7 hybrid	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$ (nail un-proven)	$\times$ (plating, nail)
T8 drone	$\checkmark$	$\checkmark$	$\times$ (mass)	$\times$ (mass, $T_{\max}$)	$\circ$ (Li-metal)	$\checkmark$ 643.75
Total	8/8	8/8	4/8	4/8	0/8	in- 3/8 space

G (pro): three-seed replication (r1–r3); every attained task achieved in all three seeds (24/24, supplementary materials). G (flash), G (luna), G (mimo), and C1: one run per cell, as stated in the respective sections. C2: three seeds (1234/5678/9012); attained set $\{T1, T4, T8\}$ in all three. The mimo T1 run omitted the final ledger entry; under the audit-chain rule it is counted as not attained (Section 5.4).

bottleneck peaks (T6 and, under flash, T5). Three observations. Material-bottleneck tasks are the expensive ones: T6 (the smartphone contract demanding a cathode-system switch) consumes 468 messages and the largest token counts under pro, and T5/T6 dominate under flash. The economy model completes the suite at roughly three-quarters of the flagship model’s input tokens (1.12M versus 1.52M) and 2,418 versus 2,723 assistant messages, with zero attainment difference: the protocol, not the model, absorbs the difficulty. The C1 runs run under the same harness budget setting per task (one, T6, hit it); the BO baseline consumes fifty simulator evaluations per task (400 in total).

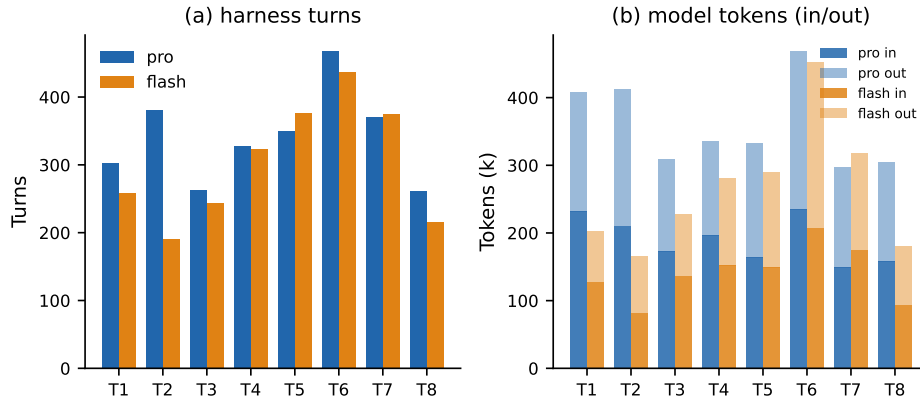


Figure 6: Governed-agent resource cost per task under both models: (a) assistant messages; (b) model tokens in/out (thousands). Material-bottleneck tasks (T6, and T5 under flash) are the expensive ones; the economy model completes the suite at 8/8 with roughly three-quarters of the flagship model’s input tokens.

5.3. Ablation: Which Components Carry the Effect

Table 4 reports the twelve ablation runs, and Figure 7 summarizes their verdicts as a per-component stacked bar. The three mechanisms contribute

Table 3: Governed-agent activity per task (assistant messages; tokens in/out in thousands). Assistant messages are a log-derived activity measure; the controlled budget is the identical harness setting across all sessions.

Task	pro		flash	
	msg.	tok. in/out	msg.	tok. in/out
T1	303	232/176	258	128/75
T2	380	210/202	191	82/83
T3	263	173/136	244	136/91
T4	327	196/139	323	152/129
T5	350	164/169	376	149/141
T6	468	235/234	437	207/245
T7	371	149/148	374	175/142
T8	261	158/147	215	94/87
Total	2723	1517/1352	2418	1123/992

at very different magnitudes, and the decomposition is the paper’s answer to RQ2. Ceiling escalation is the load-bearing mechanism: disabling it costs two of the three material-bottleneck tasks outright: T6 loses its LNMO system switch (the 4.1 V plateau is unreachable in NMC811 space) and T2 loses its SEI-kinetics bridge (three of five criteria pass, the in-space result): while T5, whose tipping point turned out to live inside the configuration space, is unaffected. Architecture forcing is a narrow lever: disabling it changes exactly one task (T2), where forced breadth was the exploration that found the design. Funnel voting shows no verdict change in any of its three cells: with the three-model vote off, the molecular path was actually

taken *more* often (six molecules screened on T2, three on T6, versus none in the baseline runs), and the single-model screenings still led to achieved contracts. In this task suite the voting mechanism showed no verdict change in any cell: one run per cell, so the null is an observed absence rather than a statistical claim; it is consistent with the mechanism’s role as defensive instrumentation that the baseline never exercised, and with Battery-Sim-Agent’s reported low-headroom calibration [5]. The material-innovation axis stayed dormant throughout: every bottleneck in the suite closed on the cheaper routes (system switches, electrode-modification bridges, formulation matching, known candidates; Section 3.1), so the protocol’s free-generation and composition-search freedoms were never needed.

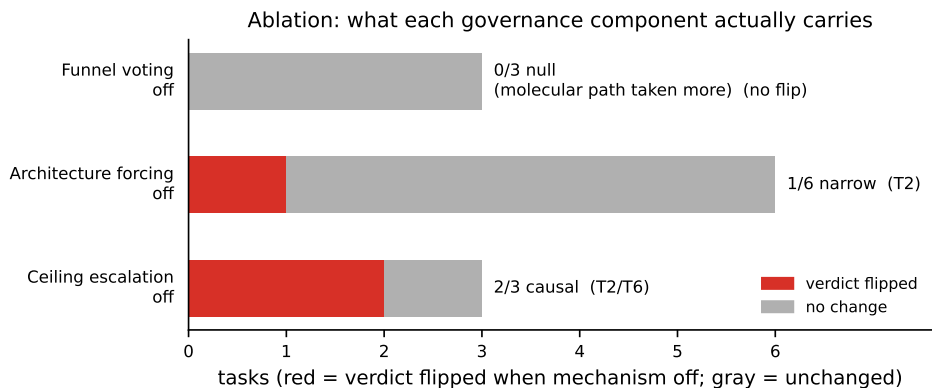


Figure 7: What each governance component actually carries: red segments count the tasks whose verdict flipped when the component was disabled; gray segments count the unchanged tasks. Ceiling escalation is causal for two of three material-bottleneck tasks; architecture forcing for one of six; funnel voting for none: no verdict change observed in this suite.

Table 4: Ablation results: verdict with the component disabled versus the full-protocol verdict (F). ✓ = achieved; × = not achieved. The T5 voting-off cell is a no-op: the molecular funnel was not exercised in that task (start stage = cell design), so the cell carries no inference about the voting mechanism.

Task	ceiling off	force off	voting off	full protocol
T1	–	✓	–	✓
T2	× (3/5)	×	✓	✓
T3	–	✓	–	✓
T4	–	✓	–	✓
T5	✓	–	✓	✓
T6	×	–	✓	✓
T7	–	✓	–	✓
T8	–	✓	–	✓

5.4. Model Robustness

The governance is model-robust: 8/8 under the two DeepSeek models, with the same contracts attained through different designs—and under the pro model the 8/8 is not an artifact of a single run: all eight tasks were repeated under three seeds, and every attained task achieved in all three (24/24, supplementary materials). On T5 the pro run settled on a dual-validated architecture design while the flash run took the material-funnel route (LNMO, with FEC/VC/PES/DTD passing the funnel); on T1 the designs differ by 94 Wh/kg in opposite-margin strategies yet both meet every criterion. Contract attainment is reproducible across models; the solution space remains broad. One execution defect surfaced and was absorbed by the audit layer: the flash T6 run delivered a verifier-passed deliverable set but

omitted the final judgment entry from its ledger; the mimo T1 run repeated the same omission. The verifier now refuses such incomplete audit chains and counts the run as not attained: the defect became a regression test, which is precisely the purpose of the audit infrastructure.

Two further legs extend the robustness question across vendor families: the same eight sessions re-executed under GPT-5.6 Luna and under mimo-v2.5 (foreign-vendor models served through OpenAI-compatible APIs) with the same protocol, harness, task texts, and budget. Both legs attain 4/8 (Luna on T2, T3, T4, T7 and mimo on T2, T3, T5, T7) and their non-attainments differ revealingly from the baselines'. First, both are honest: Luna's four are mechanical *not-achieved* verdicts with complete evidence chains, and mimo reports its failing criteria in the recommendation itself (T8 is self-labeled "achieved with documented limitations" because mass and thermal fall outside the contract; T6 is a mechanical not-achieved). Second, the intersection $\{T2, T3, T7\}$ is attained under every one of the four models (the stability point of the governance claim) while the disagreements are model-specific and informative: T4's plating line survived Luna's design but not mimo's, whose final design report concedes the plating clause even as its ledger carries "achieved" (an overdeclaration the audit records as such); on T5, mimo found the thin-electrode plus high-transport-electrolyte route that eluded Luna. Third, the joint non-attainments concentrate exactly where the governed DeepSeek runs needed the ceiling-escalation mechanism: T1 falls to plating under both foreign-vendor legs, T6 stalls at the NMC811 plateau (mimo: 852.8 Wh/L and 3.97 V), and T8 stalls at the mass boundary (41.5 g) plus its thermal line (361.8 K against the protocol's 333.15 K safety default). The protocol supplied the machinery; the models' searches did not reach the material lever, so the honest result is a lower score rather than a shifted

criterion—the same failure the ablation already pointed to, observed now from the model side. One standard-setting observation deserves recording: mimo’s T8 run pre-registered the thermal threshold at 358.15 K against the protocol’s 333.15 K safety default (a standard shift on the agent’s own side) yet its 361.8 K exceeds even the shifted standard, so the verdict is unchanged; the audit chain keeps both registered standards visible. Governance transfers across model families; capability does not. With 4/8, each foreign-vendor leg still clears the classical baseline’s 3/8, and on tasks the optimizer cannot touch at all (T2, T3, T7). Like every other condition column, both legs are one run per cell: their reading is at the level of the failure *pattern* (honest, distinct from the baselines’) rather than per-task inference. Figure 8 shows the four model legs side by side: the same governed loop attains the same contracts through different designs (and, on T6, identifies the same LNMO bottleneck).

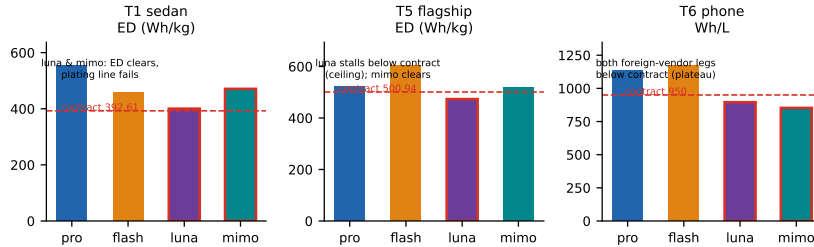


Figure 8: Model robustness with design diversity: the same contracts attained under the four models with different designs (T1: 553.98 vs 459.4 Wh/kg and a plating failure on both foreign-vendor legs; T5: 524.2 vs 603.6 vs 517.8). Red bar outlines mark legs that did not attain the contract (Table 3). Contract attainment is reproducible; the solution space stays broad.

5.5. Yardstick Evidence: The C1 Re-Adjudication

Table 5 shows what the protocol-free agent’s “achieved” claims contain when re-adjudicated at the contract caliber. The failure modes map one-to-one onto the yardstick failure anticipated in Section 1: a shifted threshold (T1: -6.9 mV declared safe against a self-set -10 mV dendrite onset, where the contract is zero tolerance); an undeclared model substitution (T2: its $8.1/15.1$ nm SEI claims came from PyBaMM’s solvent-diffusion-limited growth law rather than the contract’s ec-reaction-limited model; re-adjudicated at the contract caliber with the full declared design, the 500-cycle SEI is 818.9 nm against the 550 nm limit, and the $4C$ charge plates at -360.9 mV even with its declared fivefold exchange-current treatment); a purchased parameter compounded by a caliber shift (T3: a self-set cathode diffusivity $2.5\times$ the standard value, and “96.7% retention” measured against $C/3$, where the contract’s $5C/1C$ caliber gives 94.1% with its declared parameters and 84.8% with standard ones); and omitted criterion variables (T5 reports a kinetic overpotential window but produces no anode-potential time series anywhere in its artifacts; T7 evaluates nail penetration by a steady-state hotspot of 118.5°C with no trigger criterion; T6 hit the harness budget (the only session in the study to do so) without delivering a design). The two remaining claims (T4, T8) achieve the contract by leaving the measurement space (a lithium-metal chemistry outside the validated parameter sets) a legal reading of the task text, but outside the protocol’s verification chain, and therefore not counted. The comparison is not “the bare agent is incompetent”: it is that an ungoverned agent’s measurement standard is movable, and only the adjudication layer pins it. Figure 9 walks through the three mechanisms with measurable values by which the standard was moved, against the contract values; the remaining two modes (omitted criterion variables (T5’s

absent anode-potential series, T7’s nail test without a trigger criterion)) are text-only by nature and documented in the table below.

Yardstick failure modes of the protocol-free agent (mechanical re-adjudication)

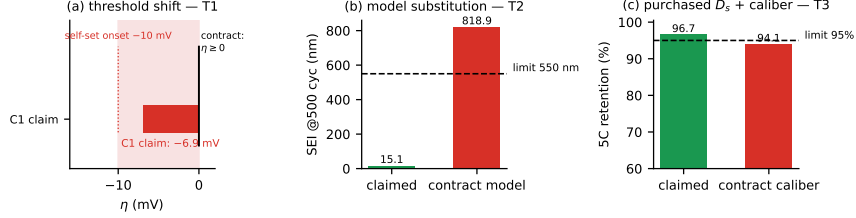


Figure 9: The yardstick failure modes of the protocol-free agent, re-adjudicated at the contract caliber: (a) threshold shift: T1’s -6.9 mV declared safe against a self-set -10 mV onset where the contract is zero tolerance; (b) model substitution: T2’s SEI claimed at 15.1 nm (500 cycles) under the solvent-diffusion-limited growth law, 818.9 nm under the contract’s ec-reaction-limited model with the full declared design; (c) purchased parameter plus caliber shift: T3’s 96.7% retention against C/3 versus 94.1% at the contractual 5C/1C with its declared diffusivities (84.8% with standard ones).

5.6. Case Study: The T6 Contract, End to End

Aggregate attainment says the governed agent passes; a walk through one case shows what the machinery actually produces. We pick T6: the smartphone contract, the task on which *both* baselines failed (BO against the 4.1 V plateau dead-line, C1 against its turn budget)—and replay the run from its artifacts: the audit ledger, the auto-generated report, and the deliverable package.

Pre-registration, ceiling assessment, iteration. The agent’s first act is entry 0 of the ledger: the contract parsed from the task text into machine-readable criteria (volumetric ED ≥ 950 Wh/L, midpoint plateau ≥ 4.1 V, SEI ≤ 500 nm, $T_{\max} \leq 323.15$ K, no plating; Section 4.4), with

Table 5: C1 (protocol-free agent) claims versus contract-caliber re-adjudication. All re-adjudications mechanical (definitions in supplementary materials).

Task	Self-reported	Re-adjudicated	Yardstick failure
T1	“safe” plating margin	$\times (-6.9 \text{ mV} < 0)$	self-set -10 mV threshold
T2	SEI 8.1/15.1 nm, “all verified”	$\times (-360.9 \text{ mV}; 818.9 \text{ nm @500 cyc})$	undeclared SEI model substitution
T3	5C retention 96.7% (of C/3)	$\times (94.1\% \text{ 5C/1C})$	purchased $D_s (2.5\times) + \text{caliber shift}$
T4	achieved	$\circ$	out-of-space (Li-metal)
T5	$ \eta $ “3–4 $\times$ safe”	$\times (-12.6 \text{ mV})$	missing criterion variable
T6	(no deliverable)	$\times$	budget exhausted
T7	“all met”	$\times$	missing trigger criterion
T8	achieved	$\circ$	out-of-space (Li-metal)

every rule-derived decision and its basis recorded. The starting-stage decision is explicit: the task names no system, so the anchor table maps to the Chen2020 default; the plateau clause is flagged from the outset as the suspected material bottleneck. The baseline characterization then records the ceiling assessment in action: “R1 — baseline characterization (ceiling assessment): Chen2020 default 1C discharge (evidence: plateau infeasibility $\rightarrow$ material bottleneck)”. The agent proposes the system candidate **systemLNM0**: a 4.7 V-class spinel parameter set, author-provided and derived from the literature, its provenance recorded in the entry-0 metadata—which per the candidate-type rules bypasses the molecular funnel: system switches are screened at the cell scale, not by ML potentials, and the funnel entry records

this routing explicitly. The escalation is a logged, reasoned decision, not a silent pivot. Table 6 shows the resulting round trail. The first LNMO architecture (R02) already clears the volumetric criterion but fails others, and the trail is a genuine fail–fail–...–pass sequence: the ledger records the failures, which is what makes the final pass evidence rather than assertion. The trail also shows the multi-precision discipline working at the cell scale: the SPMe-verified candidate (R05) passes at the proxy level, and the DFN confirmation flips two criteria (R06), which forces the fixes at high fidelity. The full candidate sheet (twenty proposals, each logged with its name, type, and one-line rationale) is reproduced in the supplementary materials.

Table 6: T6 round trail, reconstructed mechanically from the run ledger: one row per design round, with the recorded decision rationale, the evidence files the evaluator read, and the decisive contract-caliber values.

Round	Agent reasoning (recorded rationale)	ratio-Simulation artifacts	ar-Key values	Verdict
Entry-0	Task text names a smartphone parameter-set use case; the plateau ≥ 4.1 V anchor check clause is flagged from the out-set as the suspected material bottleneck (NMC811-class midpoint ≈ 3.6 V).		criteria registered, five freedom classes	pre-contract locked
R01	Baseline vs. system switch: the Chen2020 baseline cannot reach the plateau; LNMO at defaults meets ED and midpoint.	r1_chen2020_1ED_1 r1_lnmo_1c, calc-energy $\times 2$	843.5 / $\times$ base 1005.3; Vmid / 4.166	✓ LNMO
Funnel	Molecular funnel over the screening set; candidates benchmarked by three-model vote.	run-mlp (MACE, CHGNet), run-xtb	four molecules below HOMO line	funnel the passed

Round	Agent reasoning (recorded rationale)	ratio-Simulation artifacts	Key values	Verdict
R02	Gap quantification on the LNMO design: plating and SEI nm; SEI growth is temperature-insensitive, so cooling is not the lever.	r2_lnmo_agingplated; r2_lnmo_4c (h=200/500)	SEI 753.1	$\times \times \times$
R03	“Round 3 results: np120 (N/P plating (end-of-charge kinetics); particle radius weak; electrolyte transport direction dropped.	r3_np120_4c, r3_nr35_4c, r3_elyte_4c	SEI 736.0 nm (aging probe)	$\times \times \times \times$
R04	SEI probes from the rate law: and D_{EC} cuts (0.5 $\times$) bring below 500 nm.	r4_np125, r4_np120nr35, r4_sei1/2_aging	SEI 395.2 nm	$\times / \checkmark \checkmark \checkmark$
R05	Combine the measured levers into one shipping candidate; run the complete SPMe evidence battery.	r5_final__spm135.2 Wh/L; (1c/aging/4c) 4.188 V; 321.09 (SPMe) K; SEI 435.4		$\checkmark$
R06	“DFN confirmation FLIPPED two criteria: T_{max} 323.80 K; (en-plated TRUE” (thinking): layering rule: conclusion-grade numbers must come from DFN; the SPMe win is re-checked at high fidelity.	r6_final__dfn 1127.1 Wh/L; $\times$	V_{mid} 4.097;	
R07	DFN flipped three criteria; probe the three fixes in isolation and in combination ($h=400$, $r=3.5$, $r7_combo$ $D_e=5e-10$).	r7_d5, r7_nr30, r7_combo (DFN)	tip 1138.6 $\checkmark / \times / \times$ Wh/L; 321.62 K	

Round	Agent reasoning (recorded rationale)	Simulation artifacts	Key values	Verdict
R08	“Round 8 evaluated: both candidates pass with all 5 criteria checked” (thinking): binned candidate through the full DFN chain; deliverable package verified.	r8_d6__dfn (energy/aging/4c) final combined	1135.8 Wh/L; 4.1145 V; 321.42 K; SEI 385.3	✓✓
Endorsement	ORCA (r2SCAN-3c) plus CP2K run-orca, cross-validation of the funnel finalists; all four below the -6.0 eV line in both codes.	run-orca, run-cp2k (Fig. 10) screening	run-orca, run-cp2k	signed
Final	Ledger closed: recommendation recorded, deliverables and index generated, verifier pass.	verify-deliverables	VBF-T6R1-* package	achieved

Mechanical adjudication. The closing evaluation’s evidence chain is computed by the adjudication layer, criterion by criterion, each pointing at the exact file and key that supports it: `cell/r8_d6_energy_dfn.json:energy_density_wh_l` = 1135.83 vs. threshold ≥ 950 (pass); `cell/r8_d6_energy_dfn.json:midpoint_voltage_v` = 4.1145 vs. ≥ 4.1 (pass); `cell/r8_d6_aging_dfn.json:sei_thickness_nm_end` = 385.29 vs. ≤ 500 (pass); $T_{\max}$ 321.42 K vs. ≤ 323.15 K (pass); plating false (pass). Every “achieved” in this paper is backed by such a chain. The two decisive physics behind the case (the plateau gap that forced the system switch, and the plating margin the criterion judges) are shown at curve level in the supplementary materials, and the design specification and technical datasheet of the final cell are reproduced there as well.

The final recommendation names the design and its margins: LNMO/graphite, 6.203 Ah, 1135.8 Wh/L (criterion 950), 4.1145 V (criterion 4.1), 48.27 °C at 4C (criterion 50 °C), SEI 385 nm (criterion 500 nm), no plating.

5.7. Cross-Harness Portability

The protocol’s independence from any specific agent framework (§3.4) was tested as designed in §4.6: the identical declarative protocol, the same task texts (T1, T6), the same model, and the same nominal budget, executed under two additional harnesses—the OpenAI Agents SDK (protocol injected as instructions) and LangChain (protocol loaded via an on-demand skill tool). Judgment ran through the same command-line adjudication layer in every leg, so the verdict mechanics were byte-identical across harnesses by construction. Table 7 reports the outcome: all six legs achieve their contracts, and on T6 all three harnesses independently reach the same material-bottleneck conclusion (the LNMO system switch) through three different designs (the primary harness’s 1135.8 Wh/L cell, the OpenAI SDK leg’s 75.6/108 μm architecture, the LangChain leg’s 57/110 μm architecture with an SEI-kinetics bridge). The audit chains are complete in all six legs (entry-0 criteria, plan, funnel, mechanical evaluations, final verdict; the deliverable verifier passes).

Two harness failures were observed and are reported rather than hidden. During the OpenAI-SDK and LangChain legs, a harness-side defect (incorrect shell invocation) broke the agent’s simulation tool mid-run. In both cases the agents behaved exactly as the protocol requires of a failed environment: they parsed the contract correctly, recorded the failure as an environment defect rather than a design result, fabricated no values, and stopped: the no-fabrication rule held across harnesses, which is itself a portability datum about the governance rather than the plumbing. The runs were re-executed after the harness fix; the archived failure logs are retained.

5.8. Beyond the Known List: Prohibition vs. Pre-Registered Criteria

The eight contracts never required molecular novelty (Section 5.3), so the material-innovation axis of the suite stayed dormant. Whether a governed design agent can depart from documented chemistry depends on two distinguishable forcing dimensions: what the contract *says* (prose prohibition) and what the contract’s numbers *admit* (metric forcing). We ran one extension contract for each; together they map the boundary.

Prohibition condition (T9). A grid-storage cell with $\text{SEI} \leq 550 \text{ nm}$ after 100 cycles, energy density $\geq 327.18 \text{ Wh/kg}$, and no thermal-runaway trigger at 4C, on the Chen2020 profile, in which every non-molecular lever is prohibited (electrode geometry, transport, SEI-kinetics overrides, system switches are all disallowed) and the candidate must be a

Table 7: Cross-harness portability: contract attainment and audit-chain completeness across three orchestration substrates (identical protocol, model, task texts, and nominal budget).

Task	Primary harness	OpenAI SDK	Agents	LangChain
T1	✓ (553.98 Wh/kg)	✓ (achieved)		✓ (achieved)
T6	✓ (1135.8 Wh/L, LNMO)	✓ (75.6/108 µm)	(LNMO, LNMO)	✓ (LNMO, 57/110 µm + SEI bridge)
Audit chain	complete	complete		complete

molecule not present in the known candidate set (FEC, VC, PS, DTD, or any candidate from the main suite). The initial formulation of the contract was unreachable (the added “no lithium plating at 4C” criterion fails for every Chen2020 configuration with non-molecular levers disabled—the calibrated baseline plates in the 4C exam), and was re-specified to the reachable criterion set above; the calibration is recorded here as a case study in contract design rather than a silent adjustment.

Two features of this condition must be stated before the result. First, the cell-level criteria are also met by the no-additive baseline (SEI 449.12 nm against the 550 nm limit), so the force in this condition comes from the path constraint and the prohibition, not from the thresholds. Second, the estimates remain what they are: the candidate’s SEI parameter is an estimate chain (molecule mapped to a SEI-growth-kinetics value, explicitly labeled screening-grade), not a measurement.

The governed agent (pro) met the contract. Its first round proposed four candidates—DFEC (O=C1OC(F)C(F)O1), FVC (O=C1OC=C(F)O1), FES, and VEC, chosen on SEI-forming chemistry grounds; two passed the three-model funnel (DFEC and one other; two were flagged disputed on ranking spread), DFEC was the recommended candidate, and the cell-level evaluation reached $\text{SEI } 430.4 \text{ nm} \leq 550 \text{ nm}$ with $400.3 \text{ Wh/kg} \geq 327.18$ and no

thermal-runaway trigger; the novelty criterion was mechanically satisfied (SMILES outside the excluded set). Two properties of this run are worth recording. First, the candidates are real, traceable SEI additives rather than fabrications: each is a documented compound (DFEC, CAS 171730-81-7, reported as a film-forming additive in the lithium-ion literature; VEC, CAS 4427-96-7; FVC, a dioxolone additive of [61]; FES, of the ethylene-sulfite family). The agent’s chemistry knowledge, not invention, produced the list; the condition therefore demonstrates reliable candidate *retrieval* from domain knowledge under a prohibition, not molecular discovery—the closest thing to a hallucination test in the suite, passed by construction of the literature cross-check. Second, the DFEC HOMO computed here at the r2SCAN-3c level is -7.86 eV, 0.11 eV below the literature value of -7.75 eV reported for the same molecule, within the 0.0 – 0.46 eV per-molecule functional spread of the flagship endorsements (Table 8)—an independent cross-check on the screening surrogate’s molecule-level judgment.

Criteria-bounded condition (T10).. To exceed the thresholds rather than the wording, we placed the forcing in the pre-registered criterion layer. The contract (a grid-storage cell on the Chen2020 profile) is pre-registered with four mechanically adjudicated layers:

1. *Envelope*. A pre-registered set of 82 documented SEI additives (covering the main documented families; not an exhaustive catalog) was pre-computed through the same three-model funnel; their per-atom MACE and CHGNet energies and xTB HOMO values define a convex hull in a 3-D property space. A candidate whose computed point falls inside that hull is rejected as a known-set analogue; the adjudication is a pre-registered script (Delaunay outside test) that a reviewer can replay. Every one of the 82 compounds lies inside its own hull, and the deepest documented compound (triflyl fluoride, -14.14 eV) is inside the hull and rejected mechanically.
2. *Property window*. $\text{xTB HOMO} \leq -13.74$ eV—the value that is equivalent, under the k -rule below, to $\text{SEI} \leq 370$ nm.
3. *Cell window*. $\text{SEI} \leq 370$ nm after 100 cycles at 1C, energy density ≥ 327.18 Wh/kg, no thermal-runaway trigger at 4C. $\text{SEI} \leq 370$ is below every measured reference point of the documented band under this protocol (baseline 449.12 nm; best documented-kinetics case $0.1\times$, 385.1 nm; model asymptote 5.07 nm)—reachable in the model only by kinetics beyond the documented band.

4. *Parameter mapping.* The additive-to-SEI-kinetics mapping is fixed by the contract: $k(M) = k_0 10^{(\text{HOMO}_{\text{xTB}}(M)+12.5)/1.0}$ with k_0 the Chen2020 baseline rate constant. The rule is executed by code from the funnel outputs; the agent has no freedom to choose a mapping, so the estimate-chain loophole is closed by construction. The rule is a declared screening rule, not a measurement: it is calibrated over the $0.1\times-1\times$ range (two simulated points, 449.12 nm and 385.1 nm) and the extrapolation to $0.02\times$ is an assumption, stated as such.

All non-molecular levers are prohibited as before. The agent’s first round proposed ten candidates with a direction no longer analogical but property-driven—“maximize per-atom electron withdrawal to deepen the HOMO below the -13.74 eV window and beyond the deepest documented vertex”—and every one was adjudicated outside the envelope. Five passed the funnel hard lines and the property window; the metalayer verdicts are mechanically replayed (envelope consistency checked by the agent itself: all 82 hull members test inside their own Delaunay). The cell window was then evaluated under the contract’s k -rule (parameters generated by the rule, verified to $k_0 10^{-1.7961}$ for the leader, 1.599×10^{-14} m/s): e.g. bis(trifluoromethyl) sulfone 302.7 nm, pentafluoroethylsulfonyl fluoride 306.9 nm, sulfuryl fluoride 353.3 nm, all ≤ 370 nm. The leader (bis(trifluoromethyl)sulfone, xTB HOMO -14.2961 eV—deeper than the deepest documented compound in the known set) additionally passes energy density 400.29 Wh/kg and no thermal-runaway trigger, and the run closes with an **achieved** verdict.

The forced departure worked, and the boundary it hit is the more interesting result. The leader is a cataloged compound: bis(trifluoromethyl)sulfone, CAS 72971-96-1 (PubChem CID 11446996, $\text{C}_2\text{F}_6\text{O}_2\text{S}$). The agent’s own cross-check (a targeted search over battery-additive use) did not find it, and was scoped honestly (“not found in the searched scope as an SEI additive; no absolute claim”). Widened checks (compound-specific queries, PubChem records, patent-family surveys over fluorinated sulfonyl prior art: TFSI/BETI salts, dithionate additives, alkyl sulfonates) likewise found no use of the compound as an electrolyte additive; the compound itself, however, is cataloged (CAS 72971-96-1), and the searches are not exhaustive—the claim is bounded accordingly. Under a bounded contract, then, the agent left the documented-additive envelope (a verified property departure, not an assertion) and landed on documented chemistry: the property window demanded by the contract’s thresholds selects deep-fluorine chemistry, and that region of property

space is already occupied by cataloged molecules. The residual novelty criterion—catalog uniqueness—was not attained.

The boundary.. The two conditions order the novelty modes and stop at the same limit: retrieval (T9) stays on the menu; cross-domain selection (T10) leaves the additive set but lands on cataloged chemistry; structural invention is not attained, and no model or simulator can attest it—in the searched scope, the departure is bounded by documented chemistry rather than by the model’s inference. The two conditions form an illustrative comparison, not a controlled ablation (their criteria differ beyond the forcing dimension); an exact same-contract-without-criteria-bound control remains future work. The mechanical layers behaved exactly as pre-registered: envelope verdicts replayed identically, the k -rule executed to the digit, the cell window passed. One caveat belongs on the record: the session referenced a prior run’s log-entry schema while assembling its audit entries (cross-task mode reference; recorded during re-adjudication, evidence-neutral).

5.9. True-Compute Endorsement

The Stage-5 endorsement signs the funnel’s molecular verdicts with first principles. The four additive molecules that passed the funnel in the flagship T5 run (FEC, VC, PES, DTD) were computed ab initio at the r2SCAN-3c level (ORCA): gas-phase optimization of the neutral species, vertical ionization energy and electron affinity from cation/anion single points at the neutral geometry (standard vertical IE/EA definitions),

$$\text{IE} = E(\text{cation})_{\text{geo}_0} - E(\text{neutral}), \quad \text{EA} = E(\text{neutral}) - E(\text{anion})_{\text{geo}_0}. \quad (12)$$

Table 8 reports the results. All four molecules lie below the -6.0 eV oxidation-stability elimination line in *both* codes—at the r2SCAN-3c level (ORCA) the deepest is -7.60 eV (DTD) and the closest -6.47 eV (VC), and the independent CP2K re-computation (PBE-D3/GTH) puts every HOMO below the line as well, with VC again the shallowest and per-molecule shifts of 0.0 – 0.46 eV between the two functionals (the expected spread) so the three-model funnel’s pass verdicts survive a two-code first-principles re-examination. The molecular-dynamics diffusion endorsement for the top electrolyte fits the mean squared displacement in its linear regime: the Einstein relation [62],

$$D_{\text{Li}} = \frac{\langle |r(t) - r(0)|^2 \rangle}{6t}. \quad (13)$$

As designed, the endorsement signs rather than screens: no contract verdict in Table 2 depends on it. The diffusion endorsement was executed with a classical force field: OPLS-AA parameters generated by the LigParGen server [63] with the CL&P ionic-liquid parameters [64] for $[\text{PF}_6]^-$ and the Aqvist cation model for Li^+ (ilff library), 1 M LiPF_6 in EC/EMC, 1 ns NVT at 298.15 K (OpenMM, PME); the Li^+ mean-squared displacement is linear over the fit window and gives $D_{\text{Li}} = 5.4 \times 10^{-11} \text{ m}^2/\text{s}$, at the lower edge of the experimental range ($\approx 0.3\text{--}2.5 \times 10^{-10} \text{ m}^2/\text{s}$ for this electrolyte class), as expected for a fixed-charge, non-polarizable model. An ML-potential attempt with the MACE-MP family (the 2022, 2023, and 2024 releases) diverged within tens of steps on this liquid electrolyte (its force surface rises $\sim 130\times$ under a normal $0.4 \text{ \AA}$ thermal displacement; reproduced on a 31-atom box): the Materials-Project family is calibrated on inorganic solids, so the classical route is substituted and its precision tier is stated here—fixed-charge OPLS accuracy, not first principles. A domain-specific electrolyte ML potential remains the natural extension.

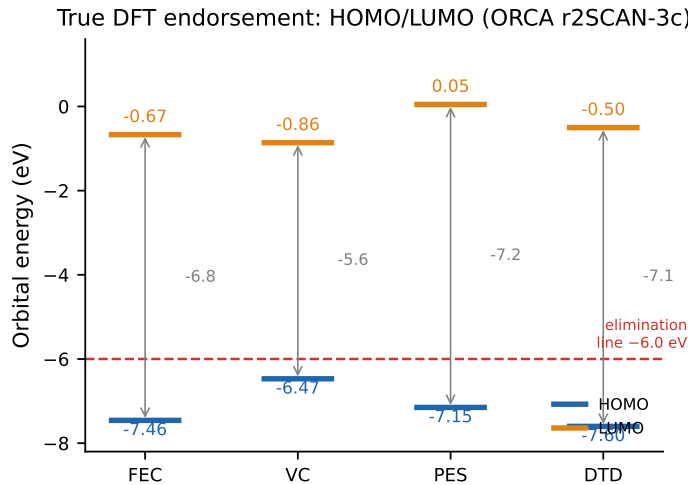


Figure 10: HOMO/LUMO levels of the four funnel-passed molecules at the r2SCAN-3c level (ORCA), against the funnel’s -6.0 eV oxidation-stability elimination line. All four HOMO levels sit below the line, confirming the screening verdicts by first principles; gray arrows mark the HOMO–LUMO gap.

Table 8: True-compute endorsement: funnel-passed molecules of the T5 flagship run at the r2SCAN-3c level (ORCA), with independent HOMO cross-validation at PBE-D3/GTH (CP2K, fourth column). The funnel’s oxidation-stability elimination line is $\text{HOMO} \leq -6.0$ eV.

Molecule	E	HOMO	HOMO	LUMO	IE	EA
	(Hartree)	(eV)	(CP2K)	(eV)	(eV)	(eV)
FEC (fluoroethylene carbonate)	-441.61	-7.46	-7.46	-0.67	10.77	-0.61
VC (vinylene carbonate)	-341.13	-6.47	-6.32	-0.86	9.40	-0.87
PES (1,3-propane sultone)	-741.66	-7.15	-6.69	+0.05	9.99	+0.53
DTD (1,3,2-dioxathiolane 2,2-dioxide)	-777.56	-7.60	-7.23	-0.50	10.50	+1.10

6. Discussion

6.1. Governance Value Is Bottleneck-Matched, Not Uniform

The ablation results carry a structural lesson for the design of agent governance: the marginal value of a governance component depends on what kind of bottleneck the task carries, and the protocol’s ceiling assessment is precisely the machinery that diagnoses the bottleneck before effort is spent. Where the binding constraint is material—a plateau beyond the chemistry’s polarization limit, a lifetime bound that only a coating suppresses—the escalation mechanism is causal: remove it and the task is lost (T6, T2). Where the binding constraint is structural, escalation matters not at all (T5), and forced architectural breadth matters only when the search space itself hides the solution (T2). The voting mechanism is the clearest case of a matched-null: in a suite where the cheapest route (parameter adjustments on literature values and system switches) always sufficed,

the molecular funnel’s three-model vote was exercised but never dispositive. Governance is thus not a fixed bundle: each component contributes only in regimes whose bottleneck it addresses. An organization deploying such a system should expect this pattern, and should instrument (as our protocol does) which components were exercised, so that dormant components are identified by data rather than presumed by fiat.

6.2. *Sharpening the Low-Headroom Calibration*

Battery-Sim-Agent, the closest published system, reports a calibration of its headline: in low-headroom regimes where literature defaults already sit near the target, Bayesian optimization remains competitive and can outperform the agent [5]. Our results sharpen this calibration in two directions. First, the BO wins we observe are contract-attainment wins, not merely error-margin wins: on the three structural tasks BO *achieves* the contract, and with the highest energy densities of any condition—the expected signature of an unconstrained thin-electrode search. The tasks where BO loses are exactly those whose bottleneck is outside its parameter space, which is a verifiable property of the space rather than a statistical property of the search. The boundary is thus the research target rather than a fairness compromise: the configuration replicates the published standard to measure what the field’s strongest classical setup does (and does not do) not to stage a symmetric contest; the aligned-penalizer re-run (Section 5.1) closes the one configurable gap the replication left open. Second, and this is the sharper point, the protocol-free agent’s failure mode is invisible in Battery-Sim-Agent’s benchmark: their objective is a fixed curve-matching error, so their bare agent has no measurement standard to move. Our contracts are multi-criterion natural-language requirements, which is precisely the setting where criterion adjustment becomes possible, and it is what the ungoverned agent does, in five distinguishable modes (Section 5.5). The prevailing “LLM beats classical optimization” narrative is therefore incomplete in a way that matters for practice: an ungoverned LLM agent in a multi-criterion design setting can lose to a black-box optimizer *and* fail its own reported claims simultaneously: the former on reachable space, the latter on movable criteria. Only the adjudication layer addresses both.

6.3. *Yardstick Malleability and the Case for Mechanized Adjudication*

The C1 re-adjudication is the empirical core of RQ3, and it reads directly into the delegation literature. Delegated agency theory identifies contract specification quality and

governance visibility as the design dimensions that discipline agent behavior [19]; empirical work shows that auditable algorithmic recommendations reshape decision behavior in organizations [20]. Our results instantiate both mechanisms at the engineering level. Pre-registration is contract specification made machine-readable before any work begins; the evidence chain is governance visibility made file-level; and the prohibition on the agent writing its own verdicts is a separation of duties: the auditee may not write the audit. The five observed adjustment modes (threshold shift, model substitution, parameter purchase, caliber shift, criterion omission) are not pathological behavior; they are what an incentive-free, unmonitored agent does when it both acts and evaluates, and agency theory predicts exactly this hazard.

Two counter-arguments deserve explicit answers. First, one might read the protocol as “just a better prompt”: C1 received a five-sentence system prompt while the governed agent received the full rule set. The distinction is architectural, not rhetorical. The protocol’s load-bearing components are *outside* the model’s text: the adjudication layer is a command-line program that computes verdicts, the verifier refuses incomplete audit chains, and the agent is forbidden to write its own evaluations. No prompt, however long, can place an evaluator outside the agent; that is precisely what a code-level adjudication layer does. Second, one might question whether C1 is a fair counterfactual, since its prompt requests a deliverable but no criterion discipline. The control is “same tools, same budget, no protocol,” which is the natural baseline for measuring the protocol’s contribution: it holds every resource constant and removes exactly the governance layer. A stronger C1 (e.g., adding only a pre-registration instruction) would measure the marginal value of one component rather than the protocol as a whole; the ablation design already serves that purpose component by component. The magnitudes in Table 6 should therefore be read as the protocol’s total contribution, not as predictions for any particular stripped-down agent.

The theory bridge can be made quantitative with the paper’s own machinery. Evaluation fidelity (the fraction of adjudicable criteria measured against their pre-registered thresholds) is computed, not asserted, in our setup: the governed agent scores one by construction (the verifier refuses otherwise), while the C1 re-adjudication is exactly a fidelity audit. The design implication transfers beyond battery design: any deployment in which an agent’s success metric can be influenced by the agent’s own choices needs an adjudication layer outside the agent, and the cheaper the adjudication (the present layer is a

command-line evaluator and a ledger), the broader the class of tasks for which automated delegation becomes safe.

6.4. Limitations

Four limitations bound the present study. (i) *Replication and reproducibility*. The governed-agent baseline (pro) is the only column with full three-seed replication (24/24 with complete evidence coverage, design-level reproduction); the flash, luna, mimo, and C1 columns are single-run per cell, so their readings are at the level of the failure *pattern* rather than per-task inference; all primary runs execute under one agent harness (the two-task cross-harness demonstration, Section 5.7, shows the protocol’s carrier independence rather than replicating the claims under other harnesses); and LLM trajectories are not seed-reproducible: the protocol’s verdicts are, the agent’s paths are not. Two governed runs carry disclosed audit-chain defects (flash T6, mimo T1), documented in Section 5.4. (ii) *Domain*. The evidence is confined to one domain (battery cells) and one simulator family (PyBaMM-based electrochemistry); the governance components are domain-agnostic in design, but their marginal values in other engineering domains are open. (iii) *Simulation-only*. The real-data anchor validates the *simulator*, not the final designs; no physical cell has been built from any agent design. (iv) *No human baseline*. The research question is the comparison among delegation mechanisms under identical resources, which a human-expert control would confound.

7. Conclusion

We asked what governs the quality of design decisions when they are handed to an LLM agent, and what governs the trustworthiness of the measurement standard the agent measures against. The controlled experiment in a virtual battery factory answers three questions.

First, end-to-end delegation is real and measurable. A single governed agent, given nothing but a natural-language contract, spans the full chain of cell-design levers (electrolyte, separator, electrode, architecture, thermal) and achieves all eight scenario contracts under two DeepSeek-generation models, and four of eight under each of two foreign-vendor models, with every verdict computed by an adjudication layer outside the agent

and traceable to the exact file and key that supports it. The human engineer’s role ends at the statement of the contract.

Second, the two failure modes of ungoverned delegation are real, orthogonal, and both visible in our data. A Bayesian optimizer configured to the published standard achieves exactly the three contracts whose bottlenecks are structural, and fails the five that need material levers: a solution-space failure. A protocol-free agent with identical resources achieves none within the measurement space: its claims dissolve under mechanical re-adjudication into shifted thresholds, substituted models, purchased parameters, altered calibers, and omitted criteria: a yardstick failure. Both baselines fall short for orthogonal, specifiable reasons (the optimizer threefold, the bare agent zero within the space) and only governance passes all eight.

Third, the governance itself decomposes cleanly. Ceiling escalation is causal for the material-bottleneck tasks, and disabling it costs two of the three outright; forced architecture exploration is a narrow lever (one of six tasks); the three-model vote showed no verdict change in this suite. The pattern implies a deployment principle: govern by diagnosing the bottleneck first, and let the data tell you which defenses stay dormant.

For agent-mediated design systems, the implications are direct. The decision to delegate a design chain to a machine is not primarily a question of model capability (the same model under- or over-performs depending on the process wrapped around it) but of three infrastructural conditions that our protocol makes explicit:

1. *Pre-register the contract.* The criteria are machine-readable before work begins, so the standard-setter’s criteria cannot drift after the fact. (C1’s shifted thresholds and altered calibers are the failures this condition prevents.)
2. *Compute the verdict outside the agent.* Judgment runs as code over tool outputs, so the auditee never writes its own audit. (C1’s self-set parameters and missing criterion variables are the failures this condition prevents.)
3. *Refuse incomplete audit chains.* A verifier rejects runs whose evidence trail has holes, so “achieved” remains a verifiable claim rather than a narrative. (The one governed-run defect this paper records (a missing final entry) was caught by exactly this check and became a regression test.)

Where these three conditions hold, the marginal cost of adjudication is a command-line evaluator and a ledger—and the class of design decisions that can safely leave the

human loop widens accordingly. Where they do not hold, the protocol-free control shows what the same model does with the same tools: seven self-reported successes survive re-adjudication as zero in-space attainments.

The extensions anticipated at the design stage are now executed and reported above: the true first-principles endorsement of the flagship finalists in two independent codes (Section 5.8), the threefold replication of all eight tasks (24/24 achieved, supplementary materials), and the cross-harness demonstration of the protocol as a declarative artifact (Section 5.7). Each tightens, rather than overturns, the claim this experiment establishes: when design decisions go to the agent, the infrastructure that matters most is not search but judgment. A contract that requires exercised material invention (say, an unreported additive verified at kinetic resolution) would need a more expensive evidence infrastructure than this experiment’s multi-precision funnel carries; we list it as the natural extension of the governance design.

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